MTH 200 Chapter 4: Applications of Derivatives

4.10 Antiderivatives Worksheet

1. Find the most general antiderivative for each of the following. Use algebra to help simplify before trying to find antiderivatives when appropriate.

a) $f(x)=10x^4-2\sec^2(x)$

$$F(x)=2x^5-2\tan(x)+C$$

b) $f(x)=\dfrac{x+2}{x}+x^{3/2}$

$$f(x)=\frac{x+2}{x}+x^{3/2}=\frac{x}{x}+\frac{2}{x}+x^{3/2}=1+2\cdot\frac{1}{x}+x^{3/2}$$
$$F(x)=x+2\ln|x|+\frac{2}{5}x^{5/2}+C$$

c) $f(x)=4\sin(x)-2x^3+\dfrac{\cos(x)}{2}$

$$F(x)=-4\cos(x)-\frac{1}{2}x^4+\frac{1}{2}\sin(x)+C$$

d) $f(x)=\dfrac{-5}{\sqrt{1-x^2}}+\dfrac{4}{\sqrt{x}}$

$$f(x)=\frac{-5}{\sqrt{1-x^2}}+\frac{4}{\sqrt{x}}=-5\cdot\frac{1}{\sqrt{1-x^2}}+4x^{-1/2}$$
$$F(x)=-5\sin^{-1}(x)+8\sqrt{x}+C$$

e) $f(x)=\dfrac{x^2-16x^3e^x+8}{8x^3}$

$$f(x)=\frac{x^2-16x^3e^x+8}{8x^3}=\frac{x^2}{8x^3}-\frac{16x^3e^x}{8x^3}+\frac{8}{8x^3}=\frac{1}{8}\cdot\frac{1}{x}-2e^x+x^{-3}$$
$$F(x)=\frac{1}{8}\ln|x|-2e^x-\frac{1}{2}x^{-2}+C$$

2. Find $g(t)$ given that $g''(t)=3t^3+3\cos(t)$, $g'(0)=7$ & $g(0)=10$.

We can see that $g'(t)=\frac{3t^4}{4}+3\sin(t)+C$ and we were told that $g'(0)=7$.

Therefore we can say that $7=\frac{3(0)^4}{4}+3\sin(0)+C \Rightarrow 7=C$ and so $g'(t)=\frac{3t^4}{4}+3\sin(t)+7$

We can then see that $g(t)=\frac{3}{4}\cdot\frac{t^5}{5}+3(-\cos(t))+7t+D=\frac{3t^5}{20}-3\cos(t)+7t+D$ and we were told that $g(0)=10$.

Therefore we can say that $10=\frac{3(0)^5}{20}-3\cos(0)+7(0)+D \Rightarrow 10=-3+D \Rightarrow D=13$

So we have that $g(t)=\frac{3t^5}{20}-3\cos(t)+7t+13$.

3. The acceleration due to gravity near the surface of some planet is $-4m/s^2$. An object is shot upward from the surface of this planet and 4 seconds later it has fallen back to the surface. What is the initial velocity of this object (be sure to include units in your answer)?

We want to find the initial velocity of the object, which $v(0)$.

We know that $s(0)=0$ & $s(4)=0$.

We calculate that:

$$a(t)=-4$$
$$v(t)=-4t+C$$
$$s(t)=-2t^2+Ct+D$$

We know that $s(0)=0$ so $0=-2(0)^2+C(0)+D \Rightarrow 0=D$ which implies that $s(t)=-2t^2+Ct+0$

We know that $s(4)=0$ so $0=-2(4)^2+C(4) \Rightarrow 0=-32+4C \Rightarrow C=8$

Therefore $s(t)=-2t^2+8t$, which means that $v(t)=-4t+8$.

We want to find the initial velocity of the object, which is $v(0)$.

$$v(0)=-4(0)+8=8m/s$$

4. A particle moves in a straight line and has acceleration given by $a(t) = 6t + 4$. Its initial velocity is $v(0) = -6m/s$ and its initial displacement is $s(0) = 9m$. Find its position function $s(t)$.

If $a(t) = 6t + 4$, then $v(t) = 3t^2 + 4t + C$.

Since we know that $v(0) = -6m/s$ we can say that $-6 = 3(0)^2 + 4(0) + C \quad \Rightarrow \quad C = -6$ which implies that $v(t) = 3t^2 + 4t - 6$.

If $v(t) = 3t^2 + 4t + 6$, then $s(t) = t^3 + 2t^2 - 6t + D$.

Since we know that $s(0) = 9m$ we can say that $9 = (0)^3 + 2(0)^2 - 6(0) + D \quad \Rightarrow \quad D = 9$ which implies that $s(t) = t^3 + 2t^2 - 6t + 9$.

5. Find *f* if $f''(x) = 12x^2 + 6x - 4, \quad f(0) = 4, \quad \& \quad f(1) = 1$.

We can see that $f'(x) = 4x^3 + 3x^2 - 4x + C$ and $f(x) = x^4 + x^3 - 2x^2 + Cx + D$.

We know that $f(0) = 4$ so, $4 = (0)^4 + (0)^3 - 2(0)^2 + C(0) + D \quad \Rightarrow \quad 4 = D$ which implies that $f(x) = x^4 + x^3 - 2x^2 + Cx + 4$

We also know that $f(1) = 1$ so, $1 = (1)^4 + (1)^3 - 2(1)^2 + C(1) + 4 \quad \Rightarrow \quad C = -3$ which implies that $f(x) = x^4 + x^3 - 2x^2 - 3x + 4$.

6. A ball is thrown upward with a speed of 48*ft/s* from the edge of a cliff 432*ft* above the ground. Find its height above the ground *t* seconds later. When does it reach its maximum height? When does it hit the ground? (Acceleration due to gravity is -32ft/s^2)

We know that $a(t) = -32$ and that $v(0) = 48$ and that $s(0) = 432$.

We can see that $v(t) = -32t + C$ and since $v(0) = 48$ we have $48 = -32(0) + C \quad \Rightarrow \quad C = 48$ which implies that $v(t) = -32t + 48$.

We can then see that $s(t) = -16t^2 + 48t + D$ and since $s(0) = 432$ we have $432 = -16(0)^2 + 48(0) + D \quad \Rightarrow \quad D = 432$ which implies that $s(t) = -16t^2 + 48t + 432$.

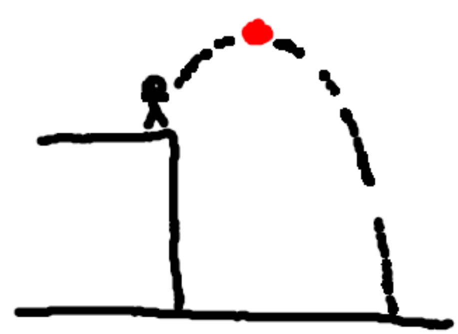
The ball reaches it maximum height when

$v(t) = 0 \quad \Rightarrow \quad 0 = -32t + 48 \quad \Rightarrow \quad t = 1.5\text{sec}$

The ball hits the ground when

$s(t) = 0 \quad \Rightarrow \quad 0 = -16t^2 + 48t + 432 \quad \Rightarrow \quad 0 = t^2 - 3t - 27$

$$\Rightarrow \quad t = \frac{3 \pm \sqrt{9 - 4(1)(-27)}}{2(1)} = \frac{3 \pm 3\sqrt{13}}{2}$$



The ball hits the ground at $t = \dfrac{3 + 3\sqrt{13}}{2} \approx 6.91\text{sec}$.